

SANKAR RAGHUTHAMAN

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EDUCATION

North Carolina State University Master of Computer Science Raleigh, NC	Aug 2024 – May 2026
Mumbai University Bachelor of Engineering, Information Technology Mumbai, India	Aug 2016 – May 2020

PROFESSIONAL EXPERIENCE

NC State University Graduate Research Intern Raleigh, NC	Sept 2025 – Mar 2026
- Built a two-phase RAG pipeline for code documentation: retrieves intent-matched examples from an open-source corpus, then enriches context with related methods from the same codebase via call graph to generate grounded comments. - Labeled 7,976 code-comment pairs by quality and intent using LLaMA-3-8B; validated against human annotations (Kappa 0.81) to ensure corpus reliability before experiments. - Intent-matched retrieval preferred by evaluators 92.2% of the time; combined pipeline with call graph context improved comment quality 90% across readability, accuracy, and grounding. Accepted at Frontiers in Education 2026.	

Children's Hospital of Philadelphia Data Science Intern Philadelphia, PA	Jun 2025 – Dec 2025
- Rebuilt genomic classification pipeline for Genetic Syndrome (~200 patients, 850K+ methylation features), fixing data leakage and replacing an ill-suited ML approach that had inflated performance estimates. - Designed nested cross-validation (5-fold outer, 3-fold inner) with all preprocessing scoped strictly inside each training fold, producing unbiased performance estimates the clinical team could trust. - Trained LinearSVC achieving 98-99% precision and 100% recall; adopted as lab standard by clinical researchers at CHOP.	

Tata Consultancy Services Mumbai, India	Aug 2020 – Jul 2024
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Machine Learning Engineer II

- Built async LLM pipeline for extracting structured fields from 1,000+ weekly invoices; validated extractions using edit distance against pre-labeled data and embedding similarity to catch field assignment errors, achieving 95% precision.
- Reduced processing time 10x (50 to 5 minutes) and human review from 100% to 5% via confidence-based routing, replacing a rule-based template system that broke silently on new formats.
- Designed reference MLOps architecture across 40+ pipelines on Azure Databricks with ML-specific CI/CD gates (schema validation, quality thresholds, drift detection), reducing pipeline failures by 80% and deployment cycles by 40%.
- Set up automated drift detection via Azure Monitor with under 5-minute alerting and MLflow model registry for versioned promotion, enabling auditable deployments across all environments.

Machine Learning Engineer

- Built real-time FX anomaly detection using Isolation Forest on Spark Structured Streaming, enabling unsupervised transaction monitoring across multiple currency pairs without labeled fraud data.
- Diagnosed and resolved production partition skew (USD/EUR at 80% volume) via composite key repartitioning, increasing throughput 25% and eliminating stale-data incidents.
- Introduced MLflow tracking and model versioning as the team's first standardized release process, bringing consistency and auditability to model deployments.

SKILLS SUMMARY

LLM and Context Engineering: RAG, GraphRAG, Multi-Agent Orchestration, Context Pipelines, Context Window Optimization, Tokenization, Few-shot Prompting, Chain-of-Thought, Conversational Memory, LLM-as-judge, Prompt Engineering, LangChain, LangGraph, ChromaDB, FAISS

Languages and Libraries: Python, SQL, Pandas, NumPy, PyTorch, scikit-learn, TensorFlow, Hugging Face Transformers, BERT, Sentence Transformers, asyncio, OpenAI API, Git

MLOps and Cloud: MLflow, Docker, Kubernetes, Azure Databricks, Azure Data Factory, Azure DevOps, Azure Monitor, GitHub Actions, CI/CD, Model Registry, Drift Detection

Data and Compute: PySpark, Spark Structured Streaming, Flask, REST APIs, Tree-sitter AST, Batch and Real-time Inference

TECHNICAL PROJECTS

Conversation Factory: Generates synthetic multi-turn tool-calling conversations for LLM training using a directed tool graph to sample structurally valid chains and multi-agent orchestration with a 3-level quality system. Diversity steering improved multi-step conversation ratio from 59% to 64%. [\[GitHub\]](#)

Contract Analyzer: Multi-agent Contract analysis platform using LangGraph with 3 parallel agents (Clause, Risk, Summarizer) for ~3x faster analysis. Multi-query RAG with citation markers highlights exact PDF passages; compliance scored against a swappable client ruleset. FastAPI + React.

RESEARCH PUBLICATIONS

- "Classification of Disease using CBC," Springer Artificial Intelligence and Soft Computing. Jul 2021
- "WIP: Usefulness-Gated, Intent-Aware Code Comment Generation with Large Language Models," Frontiers in Education 2026. [Accepted]